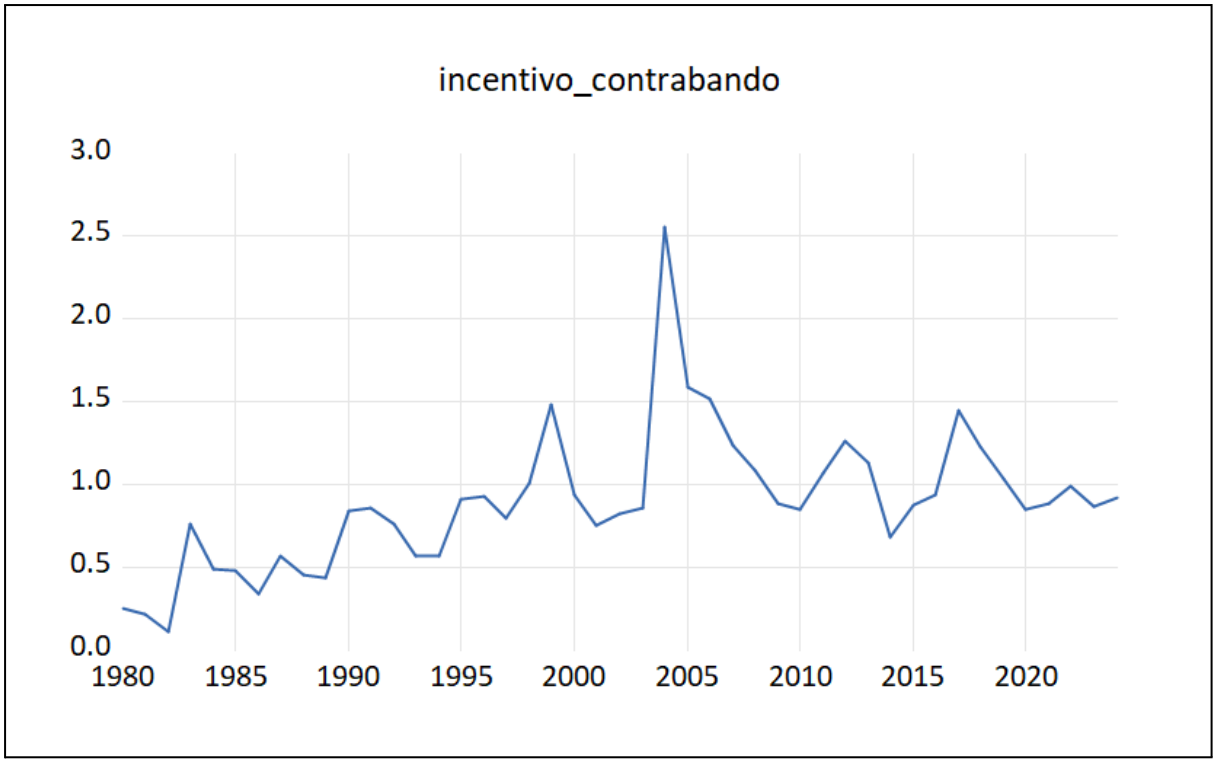


Se nota una leve tendencia entre 1980 y principios de los 2000 que termina con la presencia de un salto abrupto en el año 2004 que puede deberse a la guerra civil ocurrida en Costa de Marfil, que ocurrió entre los años 2002-2007, pero que tuvo su apogeo en el 2004. A partir del 2010 se observa cierta estabilidad.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: INCENTIVO CONTRABANDO has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.941365	0.0184
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO CONTRABANDO)
Method: Least Squares
Date: 10/14/26 Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO CONTRABANDO(-1)	-0.553339	0.140393	-3.941365	0.0003
C	0.332058	0.123311	2.692847	0.0102
@TREND("1980")	0.007811	0.004657	1.677212	0.1011

R-squared	0.277855	Mean dependent var	0.015094
Adjusted R-squared	0.242628	S.D. dependent var	0.381944
S.E. of regression	0.332394	Akaike info criterion	0.700757
Sum squared resid	4.529927	Schwarz criterion	0.822406
Log likelihood	-12.41665	Hannan-Quinn criter.	0.745870
F-statistic	7.887653	Durbin-Watson stat	2.020398
Prob(F-statistic)	0.001264		

Como la tendencia determinística no resulta significativa (p-valor = 0.1011), vuelvo a hacer el test con intercepto.

Null Hypothesis: INCENTIVO CONTRABANDO has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.525035	0.0118
Test critical values:	1% level	-3.588509	
	5% level	-2.929734	
	10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

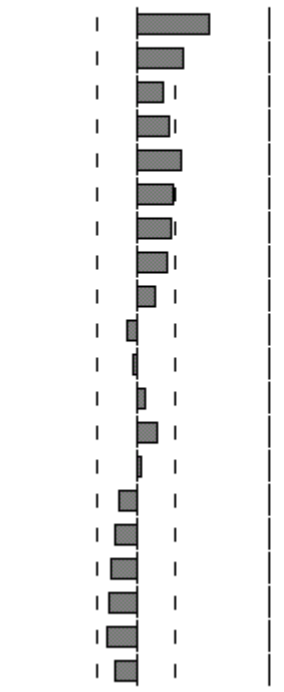
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO CONTRABANDO)
Method: Least Squares
Date: 10/14/26 Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO CONTRABANDO(-1)	-0.428289	0.121499	-3.525035	0.0010
C	0.396459	0.119683	3.312581	0.0019

R-squared	0.228308	Mean dependent var	0.015094
Adjusted R-squared	0.209935	S.D. dependent var	0.381944
S.E. of regression	0.339493	Akaike info criterion	0.721662
Sum squared resid	4.840729	Schwarz criterion	0.802761
Log likelihood	-13.87656	Hannan-Quinn criter.	0.751737
F-statistic	12.42587	Durbin-Watson stat	2.150524
Prob(F-statistic)	0.001038		

El intercepto resulta significativo por lo que analizo el p-valor asociado al estadístico tau, como 0.0118<0.05 rechazo la hipótesis nula de raíz unitaria, por lo que proceso a analizar el correlograma de la serie en niveles:

Date: 10/14/26 Time: 19:11
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.572	0.572	15.706	0.000
		2	0.373	0.069	22.546	0.000
		3	0.211	-0.039	24.795	0.000
		4	0.266	0.212	28.439	0.000
		5	0.350	0.198	34.927	0.000
		6	0.289	-0.049	39.459	0.000
		7	0.274	0.095	43.653	0.000
		8	0.251	0.088	47.244	0.000
		9	0.144	-0.162	48.462	0.000
		10	-0.075	-0.331	48.806	0.000
		11	-0.025	0.169	48.846	0.000
		12	0.071	0.109	49.166	0.000
		13	0.156	-0.065	50.780	0.000
		14	0.048	-0.097	50.935	0.000
		15	-0.126	-0.061	52.054	0.000
		16	-0.166	-0.071	54.060	0.000
		17	-0.190	-0.131	56.798	0.000
		18	-0.218	-0.090	60.527	0.000
		19	-0.226	-0.021	64.693	0.000
		20	-0.171	-0.035	67.165	0.000

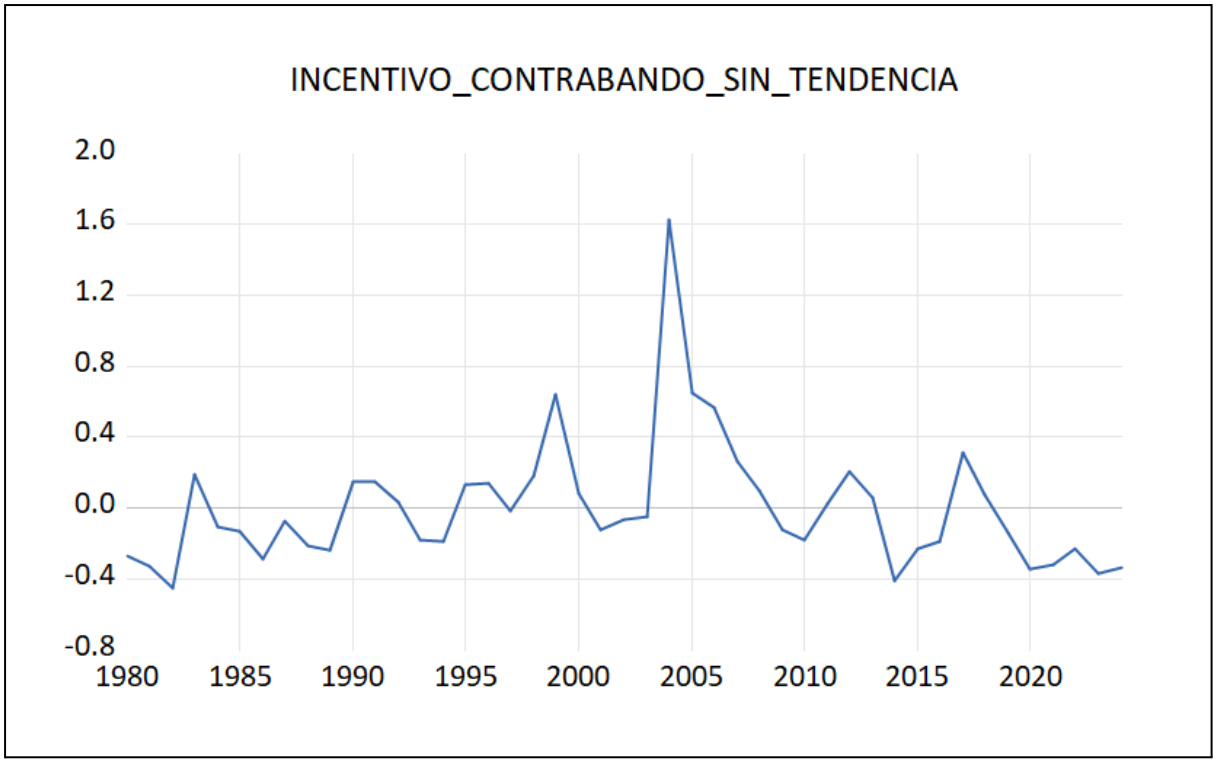
Dependent Variable: INCENTIVO CONTRABANDO
Method: Least Squares
Date: 10/14/26 Time: 19:12
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
@TREND	0.016547	0.004190	3.949091	0.0003
C	0.527008	0.107046	4.923183	0.0000

R-squared	0.266153	Mean dependent var	0.891044
Adjusted R-squared	0.249087	S.D. dependent var	0.421261
S.E. of regression	0.365045	Akaike info criterion	0.865833
Sum squared resid	5.730078	Schwarz criterion	0.946129
Log likelihood	-17.48123	Hannan-Quinn criter.	0.895766
F-statistic	15.59532	Durbin-Watson stat	1.094744
Prob(F-statistic)	0.000286		

Notar que en el ADF la tendencia no resultaba significativa, sin embargo al estimarla junto al intercepto tiene un p-valor = 0.0003.

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:



Date: 10/14/26 Time: 19:23
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.436	0.436	9.1448	0.002
		2	0.186	-0.005	10.854	0.004
		3	0.005	-0.093	10.855	0.013
		4	0.070	0.126	11.105	0.025
		5	0.192	0.168	13.048	0.023
		6	0.130	-0.050	13.958	0.030
		7	0.155	0.110	15.289	0.032
		8	0.170	0.123	16.937	0.031
		9	0.049	-0.130	17.077	0.048
		10	-0.227	-0.338	20.199	0.027
		11	-0.174	0.093	22.082	0.024
		12	-0.030	0.094	22.140	0.036
		13	0.117	-0.015	23.048	0.041
		14	0.002	-0.128	23.048	0.059
		15	-0.215	-0.136	26.320	0.035
		16	-0.269	-0.114	31.598	0.011
		17	-0.288	-0.144	37.884	0.003
		18	-0.293	-0.122	44.591	0.000
		19	-0.263	-0.057	50.210	0.000
		20	-0.171	-0.085	52.686	0.000

Analizando el correlograma, tanto la FAC como la FACP resultan significativas solamente en el primer rezago (y el rezago 10 de la FACP)

Dependent Variable: INCENTIVO_CONTRABANDO_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/20/26 Time: 18:28
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Failure to improve likelihood (non-zero gradients) after 8 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.491736	0.295234	1.665583	0.1032
MA(1)	-0.060339	0.338114	-0.178459	0.8592

R-squared	0.197069	Mean dependent var	0.006215
Adjusted R-squared	0.177951	S.D. dependent var	0.362601
S.E. of regression	0.328759	Akaike info criterion	0.657404
Sum squared resid	4.539457	Schwarz criterion	0.738504
Log likelihood	-12.46289	Hannan-Quinn criter.	0.687480
Durbin-Watson stat	1.982554		

Inverted AR Roots	.49
Inverted MA Roots	.06

Si se estima un ARMA(1,1) ninguno de los dos coeficientes resulta significativo.

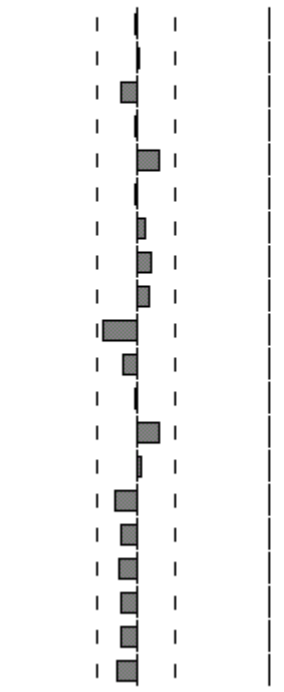
Dependent Variable: INCENTIVO_CONTRABANDO_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:26
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 3 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.445011	0.137157	3.244539	0.0023

R-squared	0.196426	Mean dependent var	0.006215
Adjusted R-squared	0.196426	S.D. dependent var	0.362601
S.E. of regression	0.325043	Akaike info criterion	0.612749
Sum squared resid	4.543090	Schwarz criterion	0.653299
Log likelihood	-12.48049	Hannan-Quinn criter.	0.627787
Durbin-Watson stat	2.011150		

Inverted AR Roots	.45

Date: 10/14/26 Time: 19:27
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term



	AC	PAC	Q-Stat	Prob
1	-0.014	-0.014	0.0089	
2	0.024	0.024	0.0369	0.848
3	-0.111	-0.110	0.6450	0.724
4	-0.012	-0.015	0.6520	0.884
5	0.181	0.189	2.3583	0.670
6	-0.008	-0.016	2.3617	0.797
7	0.070	0.057	2.6294	0.854
8	0.118	0.170	3.4088	0.845
9	0.099	0.107	3.9747	0.859
10	-0.257	-0.309	7.9165	0.543
11	-0.100	-0.089	8.5357	0.577
12	-0.015	0.024	8.5507	0.663
13	0.173	0.077	10.514	0.571
14	0.046	-0.032	10.658	0.639
15	-0.159	-0.089	12.431	0.572
16	-0.115	-0.101	13.385	0.573
17	-0.126	-0.132	14.574	0.556
18	-0.123	-0.153	15.747	0.542
19	-0.113	-0.073	16.782	0.538
20	-0.141	-0.218	18.454	0.492

Estimando solamente el AR(1) arroja coeficiente significativo, residuos ruido blanco y raíz dentro del círculo unitario, sin embargo debo probar como se comporta el componente MA(1) sin el AR(1):

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:28
Sample: 1980 2024
Included observations: 45
Failure to improve likelihood (non-zero gradients) after 12 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1979

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.368432	0.139890	2.633723	0.0116

R-squared	0.160268	Mean dependent var	1.78E-16
Adjusted R-squared	0.160268	S.D. dependent var	0.360873
S.E. of regression	0.330692	Akaike info criterion	0.646716
Sum squared resid	4.811730	Schwarz criterion	0.686864
Log likelihood	-13.55110	Hannan-Quinn criter.	0.661682
Durbin-Watson stat	1.833159		

Inverted MA Roots	-.37

Date: 10/14/26 Time: 19:28
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.072	0.072	0.2512	
		2	0.197	0.193	2.1649	0.141
		3	-0.068	-0.097	2.3957	0.302
		4	0.041	0.015	2.4845	0.478
		5	0.179	0.218	4.1759	0.383
		6	0.042	-0.006	4.2711	0.511
		7	0.114	0.039	4.9921	0.545
		8	0.105	0.139	5.6178	0.585
		9	0.096	0.050	6.1643	0.629
		10	-0.238	-0.358	9.5844	0.385
		11	-0.082	-0.063	10.007	0.440
		12	-0.047	0.107	10.146	0.517
		13	0.130	0.046	11.256	0.507
		14	0.016	-0.082	11.272	0.588
		15	-0.174	-0.129	13.416	0.494
		16	-0.161	-0.116	15.314	0.429
		17	-0.187	-0.133	17.962	0.326
		18	-0.196	-0.164	20.980	0.227
		19	-0.160	-0.042	23.062	0.188
		20	-0.173	-0.167	25.599	0.142

El coeficiente MA(1) por separado arroja el problema de “Failure to improve likelihood…”, lo cual puede deberse a la poca cantidad de observaciones, por lo que este modelo podría ser problemático.

Sin embargo si se pasa por alto eso y se quiere comparar ambos modelos, necesito comparar con el criterio de información Akaike, el cual requiere que ambos modelos tengan la misma cantidad de observaciones, por lo que estimo nuevamente el MA(1) con 44 observaciones:

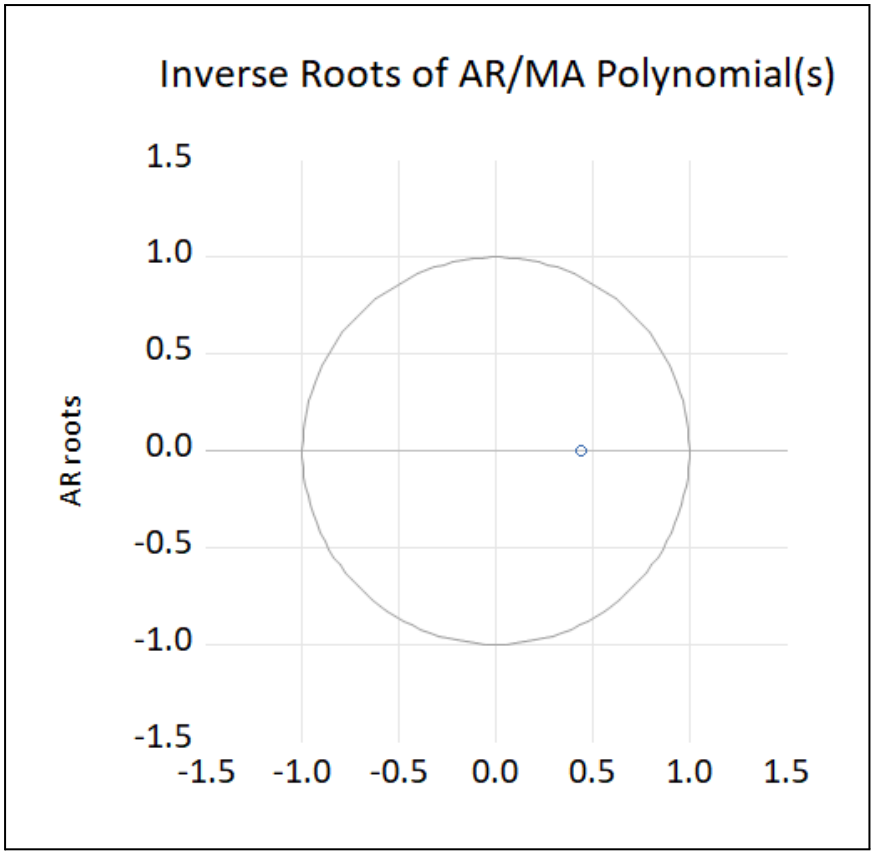
Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:29
Sample: 1981 2024
Included observations: 44
Failure to improve likelihood (non-zero gradients) after 12 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

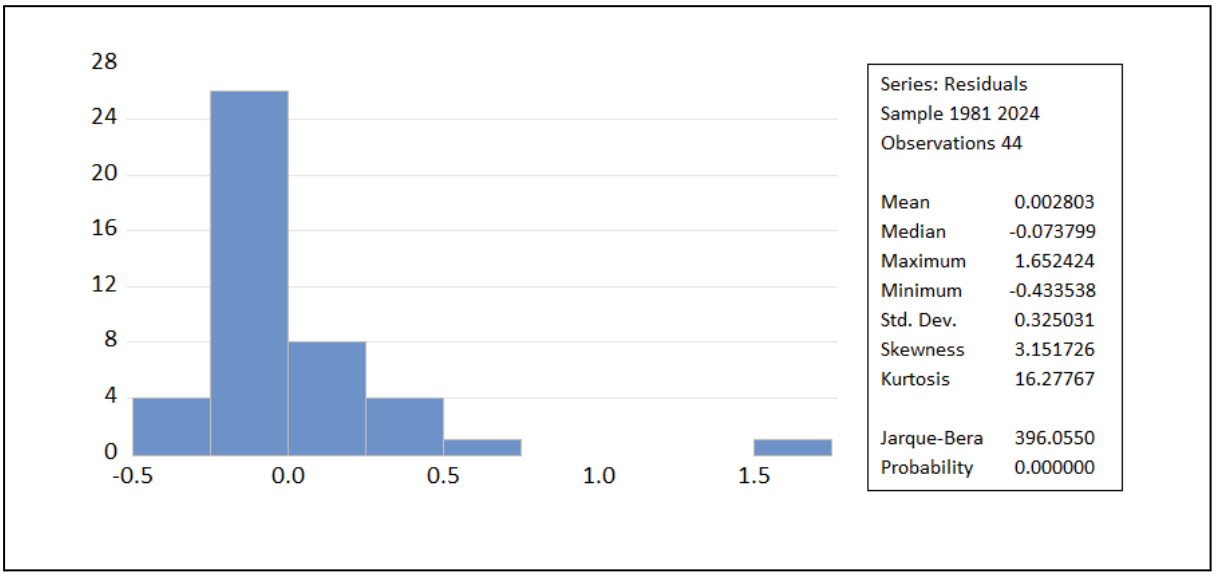
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.367926	0.142016	2.590728	0.0130

R-squared	0.157091	Mean dependent var	0.006215
Adjusted R-squared	0.157091	S.D. dependent var	0.362601
S.E. of regression	0.332904	Akaike info criterion	0.660539
Sum squared resid	4.765476	Schwarz criterion	0.701089
Log likelihood	-13.53187	Hannan-Quinn criter.	0.675577
Durbin-Watson stat	1.832781		

Inverted MA Roots	-.37

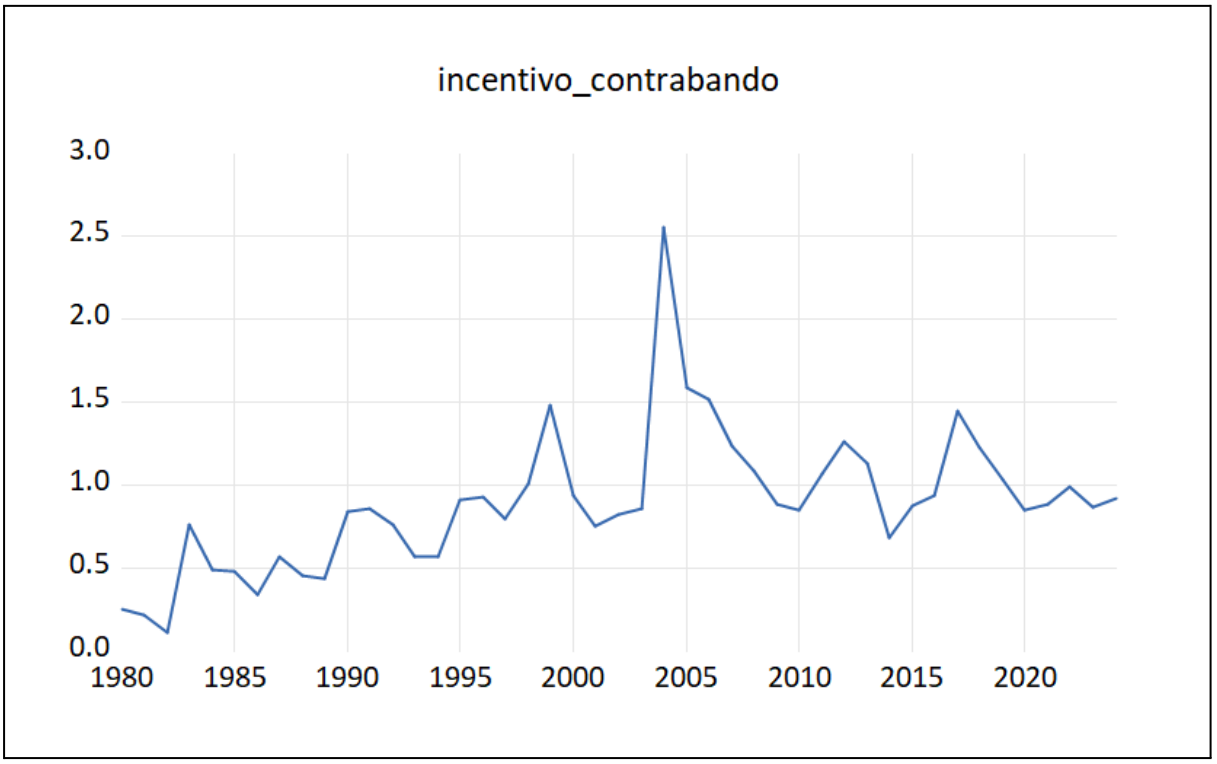
La nueva estimación del MA(1) sigue arrojando el problema de convergencia, y además tiene un peor AIC que el AR(1), 0.66>0.61, por lo que siguiendo este criterio debería quedarme con el modelo AR(1).





El test de Jarque-Bera del modelo AR(1) rechaza la hipótesis nula de que los residuos se comportan normalmente, por una fuerte asimetría positiva (3,151), y una curtosis elevada (16.27), **lo cual puede deberse al valor atípico del año 2004.**

Incorporando una Variable de intervención:



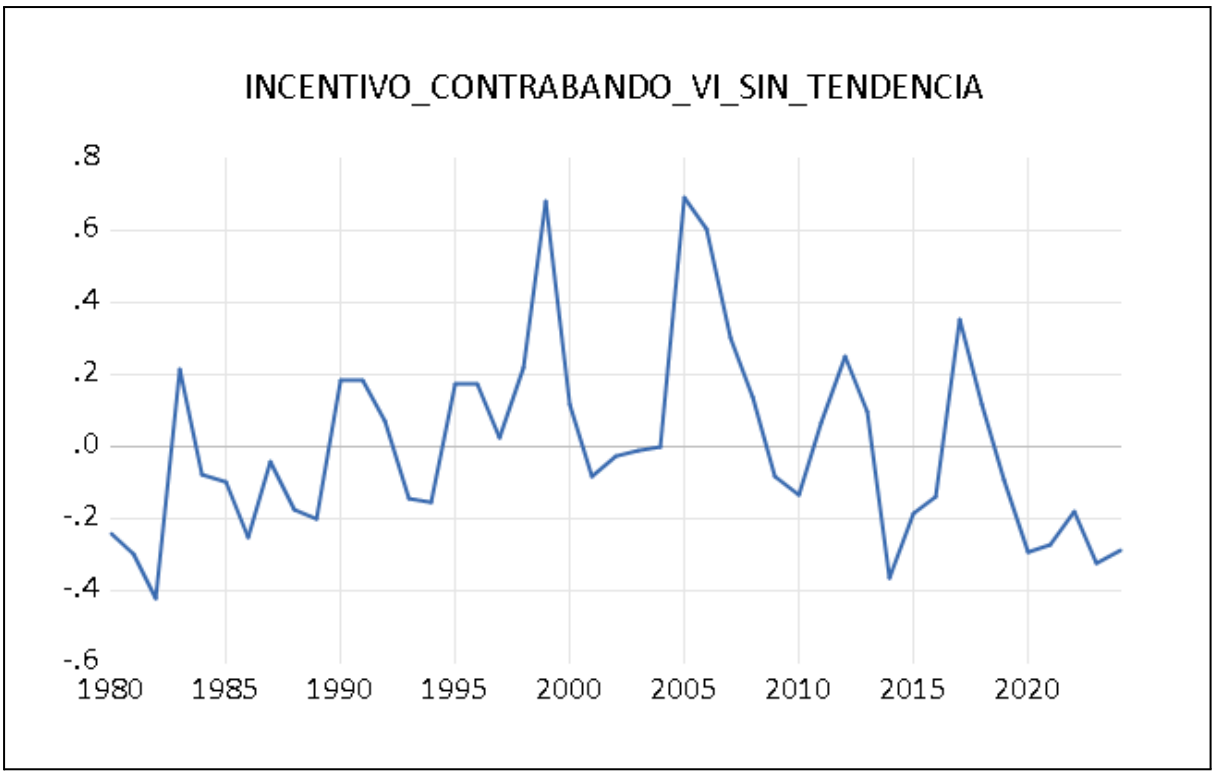
En este caso la variable de intervención se incorpora en el año 2004, debido a que fue el pico de la guerra civil de costa de marfil, que duró desde 2002 hasta 2007.
En ese año el precio de productor en costa de marfil fue extremadamente bajo, comparado con el de Ghana, lo que provoca que el incentivo de contrabando sea tan elevado.

Dependent Variable: INCENTIVO_CONTRABANDO
Method: Least Squares
Date: 11/02/26 Time: 13:16
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.499616	0.078658	6.351742	0.0000
@TREND	0.016108	0.003075	5.238586	0.0000
VI_2004	1.667653	0.270907	6.155822	0.0000

R-squared	0.614220	Mean dependent var	0.891044
Adjusted R-squared	0.595849	S.D. dependent var	0.421261
S.E. of regression	0.267808	Akaike info criterion	0.267244
Sum squared resid	3.012277	Schwarz criterion	0.387688
Log likelihood	-3.012993	Hannan-Quinn criter.	0.312144
F-statistic	33.43515	Durbin-Watson stat	0.984937
Prob(F-statistic)	0.000000		

La variable de impacto, junto con la tendencia y el intercepto resultó significativa



Null Hypothesis: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.778555	0.0003
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

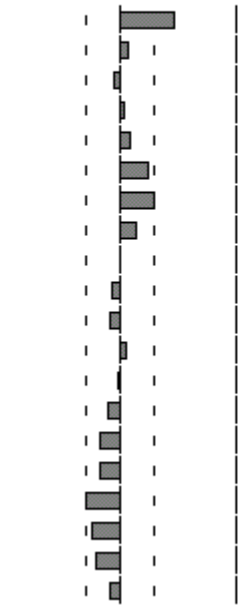
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
)
Method: Least Squares
Date: 11/02/26 Time: 13:18
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO_CONTRABANDO_VI_SIN_TE	-0.502591	0.133011	-3.778555	0.0005

R-squared	0.249257	Mean dependent var	-0.001013
Adjusted R-squared	0.249257	S.D. dependent var	0.262672
S.E. of regression	0.227593	Akaike info criterion	-0.100047
Sum squared resid	2.227348	Schwarz criterion	-0.059497
Log likelihood	3.201025	Hannan-Quinn criter.	-0.085009
Durbin-Watson stat	1.814054		

Date: 11/02/26 Time: 13:18
Sample: 1980 2024
Included observations: 45



	AC	PAC	Q-Stat	Prob
1	0.483	0.483	11.235	0.001
2	0.074	-0.209	11.502	0.003
3	-0.066	-0.010	11.719	0.008
4	0.038	0.131	11.793	0.019
5	0.088	-0.003	12.203	0.032
6	0.256	0.274	15.758	0.015
7	0.302	0.097	20.824	0.004
8	0.141	-0.075	21.960	0.005
9	0.002	0.041	21.960	0.009
10	-0.081	-0.116	22.358	0.013
11	-0.082	-0.052	22.778	0.019
12	0.047	0.104	22.921	0.028
13	-0.015	-0.281	22.935	0.042
14	-0.102	-0.044	23.649	0.050
15	-0.181	-0.103	25.955	0.038
16	-0.185	-0.153	28.448	0.028
17	-0.312	-0.202	35.784	0.005
18	-0.258	-0.075	40.984	0.002
19	-0.219	-0.208	44.876	0.001
20	-0.092	0.147	45.592	0.001

Dependent Variable: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:19
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 3 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.497409	0.133011	3.739602	0.0005

R-squared	0.245059	Mean dependent var	0.005592
Adjusted R-squared	0.245059	S.D. dependent var	0.261941
S.E. of regression	0.227593	Akaike info criterion	-0.100047
Sum squared resid	2.227348	Schwarz criterion	-0.059497
Log likelihood	3.201025	Hannan-Quinn criter.	-0.085009
Durbin-Watson stat	1.814054		

Inverted AR Roots	.50

Date: 11/02/26 Time: 13:20
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.082	0.082	0.3154	
		2	-0.173	-0.181	1.7550	0.185
		3	-0.136	-0.109	2.6723	0.263
		4	0.036	0.027	2.7362	0.434
		5	-0.041	-0.092	2.8229	0.588
		6	0.155	0.171	4.1060	0.534
		7	0.253	0.231	7.6212	0.267
		8	-0.012	-0.015	7.6297	0.366
		9	-0.049	0.090	7.7662	0.457
		10	-0.052	-0.021	7.9283	0.541
		11	-0.080	-0.103	8.3241	0.597
		12	0.143	0.181	9.6105	0.566
		13	-0.023	-0.197	9.6463	0.647
		14	-0.061	-0.080	9.8933	0.703
		15	-0.080	-0.043	10.337	0.737
		16	0.038	-0.067	10.439	0.791
		17	-0.236	-0.242	14.605	0.554
		18	-0.056	-0.032	14.849	0.606
		19	-0.088	-0.295	15.480	0.629
		20	-0.042	-0.050	15.629	0.682

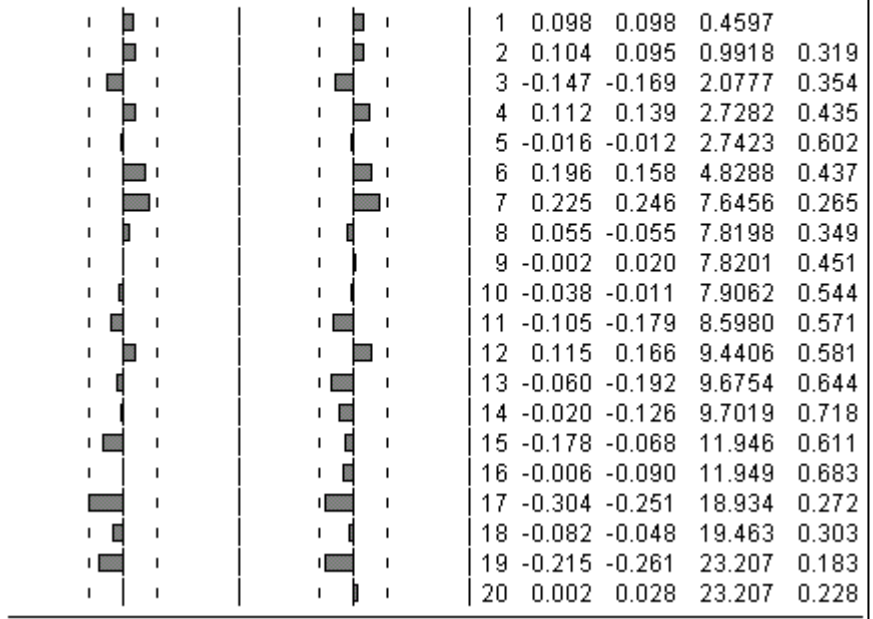
Dependent Variable: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/02/26 Time: 13:20
Sample: 1980 2024
Included observations: 45
Convergence achieved after 42 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.478994	0.150264	3.187685	0.0027
SIGMASQ	0.050568	0.008774	5.763403	0.0000

R-squared	0.244574	Mean dependent var	-5.43E-17
Adjusted R-squared	0.227006	S.D. dependent var	0.261650
S.E. of regression	0.230043	Akaike info criterion	-0.051882
Sum squared resid	2.275551	Schwarz criterion	0.028414
Log likelihood	3.167355	Hannan-Quinn criter.	-0.021949
Durbin-Watson stat	1.771654		

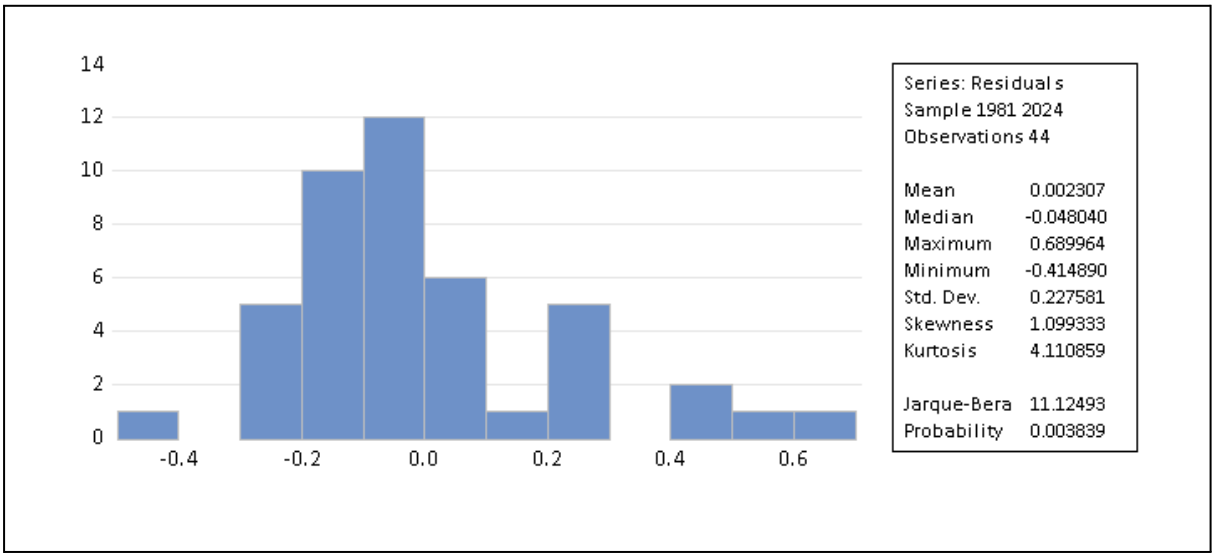
Inverted MA Roots	-.48

Date: 11/02/26 Time: 13:21
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term



	Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
1			0.098	0.098	0.4597	
2			0.104	0.095	0.9918	0.319
3			-0.147	-0.169	2.0777	0.354
4			0.112	0.139	2.7282	0.435
5			-0.016	-0.012	2.7423	0.602
6			0.196	0.158	4.8288	0.437
7			0.225	0.246	7.6456	0.265
8			0.055	-0.055	7.8198	0.349
9			-0.002	0.020	7.8201	0.451
10			-0.038	-0.011	7.9062	0.544
11			-0.105	-0.179	8.5980	0.571
12			0.115	0.166	9.4406	0.581
13			-0.060	-0.192	9.6754	0.644
14			-0.020	-0.126	9.7019	0.718
15			-0.178	-0.068	11.946	0.611
16			-0.006	-0.090	11.949	0.683
17			-0.304	-0.251	18.934	0.272
18			-0.082	-0.048	19.463	0.303
19			-0.215	-0.261	23.207	0.183
20			0.002	0.028	23.207	0.228

Comparo ambos modelos según el criterio de Akaike, en donde gana el modelo con AR(1) (-0.1<-0.05)



Comparado con el modelo sin variable de intervención se ve que mejora mucho el criterio de información y disminuye la varianza, además a pesar de que el test de normalidad sigue siendo rechazado tiene mucha menor asimetría y curtosis que el modelo anterior.